

Jose Barbeito

Written Summary on Glossina Tachinoides, the Tsetse fly

11/7/23

Glossina Tachinoides is a species of Tsetse fly. The Tsetse fly lives in sub-Saharan Africa and is a blood feeding insect, feeding primarily on wild hogs, domestic cattle, humans, and other mammals. During my research, I was shocked to find the wide variety of sources of blood for this little fly. With around 25 different species, a wide variety of hosts have been found for this fly, including warthogs, giraffe, hippopotamus's, buffalo and kudu, armadillos and porcupines, crocodiles, hundreds of species of birds, rhinos, snakes, elephants, ostriches, and of course, humans. The wide variety of food sources leads to a worrying source of disease, since many Glossina flies, but especially Tachinoides, carry the parasite responsible for African Sleeping Sickness, a trypanosomiasis parasite. This parasite lives in the liver, heart, and kidney of mammals and can spread to the brain, causing coma and death.

The African variant is aptly named Sleeping Sickness due to this coma-inducing parasite, with around 700 cases being reported in 2022. The number of infections, while low today, has dropped significantly since 1999, where around 28,000 cases were reported. There could be many causes to this, but awareness and proper prevention is most likely the cause.

The life cycle of a Tsetse fly is peculiar and noteworthy. The mother tsetse flies begins by producing an egg inside her uterus, which grows over the course of 10 days, going through the three larval instar stages. The offspring are fed a milky substance created by the mother. Because of this, the mother is a voracious feeder during pregnancy, because she has to feed not only herself but the offspring she carries. At the end of the third instar, the larva crawls out of the mother. At this point, the larva is almost the size of the mother, leading to a scenic birth. The larva, once outside, will harden its shell and complete its metamorphosis. The mother gives birth every 10 days, leading to a slow and linear growth curve. This is quite opposite to normal flies, who lay many eggs on a substrate and leave them to fend for themselves.

I found many other interesting facts about the Tsetse fly while doing my research for the presentation. Along with its spectacular reproductive cycle, the Glossina Tachinoides has been targeted for control in livestock, primarily because of its many disease vectors, but specifically sleeping sickness. In animals, sleeping sickness is called Nagana. Ancient control techniques include clearing land, slaughtering wild animals, and pesticides, all of which are ecologically disastrous. More recent control techniques include direct trapping of the flies and sterilization of mass produced males, which are then released into the environment. Both methods have had success, and are being implemented in cattle and livestock farms in Africa today. To trap the flies, scientists and farmers use chemical attractants, including acetone, carbon dioxide, and octenol. Cheaper attractants are usually used for mass trapping, the primary source of which is cow urine. Lastly, another fun fact about the fly is that there is a theory that the stripes on Zebras are meant to confuse Tsetse flies, not for camouflage, as black and white alternating bands have been shown to be confusing for the flies, who prefer darker hide such as those of cows and buffalo.

In conclusion, the Tsetse fly is an annoying, disease driving blood feeder in Africa that feeds on a wide variety of animals, yet its unique reproductive cycle makes it deserving of a closer look.

Sources

1. <https://www.who.int/data/gho/data/themes/topics/human-african-trypanosomiasis#:~:text=Between%201999%20and%202022%2C%20the,from%2027%20862%20to%20799.>
2. https://en.wikipedia.org/wiki/Tsetse_fly
3. https://en.wikipedia.org/wiki/Tsetse_fly
4. <https://bmresnotes.biomedcentral.com/articles/10.1186/s13104-020-04970-1>
5. https://www.google.com/search?sca_esv=580187299&rlz=1C1CHBF_enUS1057US1057&q=tsetse+fly&tbm=isch&source=lnms&sa=X&ved=2ahUKewiimteuu7KCAxUpM1kFHaE6C1QQ0pQJegQIDhAB&biw=1920&bih=931&dpr=1#imgsrc=CwzTbNyH-uL_BM
6. <https://www.sciencedirect.com/science/article/abs/pii/S0001706X10001750>
7. <https://www.britannica.com/animal/tsetse-fly>
8. <https://www.britannica.com/science/mycosis>
9. <https://wwwnc.cdc.gov/travel/diseases/african-sleeping-sickness-african-trypanosomiasis#:~:text=African%20trypanosomiasis%2C%20also%20called%20sleeping,treated%2C%20it%20can%20cause%20death.>
10. <https://www.advancedsciencenews.com/meet-the-tsetse-fly-the-supremom-of-the-insect-world/>
- 11.